Concepts beyond the lab syllabus

Every module and library used by Meridian that is **not** covered by the Software Lab-1 practicals has its own explainer here. Each one is written for a reader who has done Labs 1 to 6 and nothing more.

The rule the project follows: if a line of code needs a concept that never appeared in a practical, that concept gets a file in this folder.

The list

File	Concept	Where it is used	Closest lab
flask.md	Flask routing, `jsonify`, `request`	`backend/app.py`	Lab 4, functions
csv-module.md	`csv.DictReader`, `csv.DictWriter`	`backend/store.py`	Lab 4, file handling
shutil-and-zipfile.md	`shutil.copyfile`, `zipfile.ZipFile`	`backend/store.py`, `scripts/fetch_datasets.py`	Lab 4, `os` module
pandas-select-dtypes.md	`select_dtypes(include=...)`	`backend/analytics.py`, `scripts/profile_dataset.py`	Lab 6, pandas
pandas-missing-values.md	`na_values`, `keep_default_na`	`backend/analytics.py`, `scripts/profile_dataset.py`	Lab 6, pandas
matplotlib-basics.md	`pyplot`, figures, `savefig`	`backend/analytics.py`	Lab 5, matplotlib
requests.md	`requests.get`, status codes	`scripts/fetch_datasets.py`	syllabus, web scraping
json.md	`json.dumps`, `json.loads`, `allow_nan`	`backend/analytics.py`, `tests/`	Lab 4, file handling

Concepts deliberately NOT used

Worth recording, because a reviewer may ask why these are absent:

Concept	Why not
`bs4` / BeautifulSoup	The syllabus pairs it with `requests` for scraping HTML. Our goal was structured CSV files, not HTML pages, so parsing markup would have been the wrong tool. `requests.md` explains this.
`os.walk`	Used in an early version of `backup_all`. Removed: it swept `data/datasets/` too, where two files share a name with registry files and silently overwrote one another when flattened. See `shutil-and-zipfile.md`.
pytest	Not in any practical. Tests are plain functions with `assert`, run by `tests/run_tests.py`.
Any database	CSV only, by design. The brief called for file-based storage.
`scipy`	Listed in the syllabus but never needed. Everything numeric was done with NumPy and pandas.

Verifying this list is complete

Run:

```
python scripts/check_concept_coverage.py
```

It parses every `import` statement in `backend/` and `scripts/`, and reports any module that is neither a standard library module explained above nor covered by one of these files.